CHARACTER MANIPULATION

Subroutine to Input 2 Characters and pack into a word

```
IN2,    --              / Subroutine entry
FST,    SKI
        BUN  FST
        INP             / Input 1st character
        OUT
        BSA  SH4        / Logical Shift left 4 bits
        BSA  SH4        / 4 more bits
SCD,    SKI
        BUN  SCD
        INP             / Input 2nd character
        OUT
        BUN  IN2  I     / Return
```

Program to store input characters in a buffer

```
        LDA  ADS        / Load 1st address of buffer
        STA  PTR        / Initialize pointer
LOP,    BSA IN2         / Call IN2
        STA  PTR  I     / Str double char. W. in buffer
        ISZ  PTR        / Inc Pointer
        BUN  LOP        / BUN to input more char.
        HLT
ADS,    HEX  500        / First address of buffer
PTR,    HEX  0          / Location for pointer
```

Program to Compare two words

```
        LDA WDI         /Load first word
        CMA
        INC             /Form 2's complement
        ADD WD2         /Add second word
        SZA             /Skip if AC is zero
        BUN UEQ         /Branch to "unequal" routine
        BUN EQL         /Branch to equal routine
WD1,    ____
WD2,    ____
```

PROGRAM INTERRUPT

Tasks of Interrupt Service Routine

- Save the Status of CPU
  Contents of processor registers and Flags

- Identify the source of Interrupt
  Check which flag is set

- Service the device whose flag is set
  (Input Output Subroutine)

- Restore contents of processor registers and flags

- Turn the interrupt facility on

- Return to the running program
  Load PC of the interrupted program

INTERRUPT SERVICE ROUTINE

```
Loc.
0      ZRO,   -              / Return address stored here
1             BUN  SRV       / Branch to service routine
100           CLA            / Portion of running program
101           ION            / Turn on interrupt facility
102           LDA  X
103           ADD  Y         / Interrupt occurs here
104           STA  Z         / Program returns here after interrupt

                             / Interrupt service routine
200    SRV,   STA  SAC       / Store content of AC
              CIR            / Move E into AC(1)
              STA  SE        / Store content of E
              SKI            / Check input flag
              BUN  NXT       / Flag is off, check next flag
              INP            / Flag is on, input character
              OUT            / Print character
              STA  PT1  I    / Store it in input buffer
              ISZ  PT1       / Increment input pointer
       NXT,   SKO            / Check output flag
              BUN  EXT       / Flag is off, exit
              LDA  PT2  I    / Load character from output buffer
              OUT            / Output character
              ISZ  PT2       / Increment output pointer
       EXT,   LDA  SE        / Restore value of AC(1)
              CIL            / Shift it to E
              LDA  SAC       / Restore content of AC
              ION            / Turn interrupt on
              BUN  ZRO  I    / Return to running program
       SAC,   -              / AC is stored here
       SE,    -              / E is stored here
       PT1,   -              / Pointer of input buffer
       PT2,   -              / Pointer of output buffer
```